Multimodal Graph Learning Framework for Mortality and Sepsis Prediction Using eICU

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Introduction

Intensive care units (ICU) are specialized departments within hospitals that provide continuous care and high-level medical intervention for critically ill patients. Predicting the early mortality and sepsis is crucial for interventions, better clinical decisions, and optimal use of resources. Traditional severity scores are not able to capture the complex conditions. deep learning models can capture patterns in large quantity of clinical data typically recorded by EHR systems,

This work investigates different approaches to predict in-hospital mortality and sepsis based on eICU collaborative research database (eICU-CRD). There are two modalities present: static and time series based. LSTM and GRU are implemented to generate temporal embeddings and a denoising autoencoder for static data. Instead of treating each patient as a separate individual vector, several patients with similar conditions are grouped together and a GCN used to generate predictions on mortality and sepsis.

Background and Literature Review

Research	Architecture	Dataset/Objective
Awad et al. (2020) [1]	Time series analysis and Data mining models,	ICU mortality using first 48 hours
Wang et al. (2024) [2]	MFNet heirarchial fusion (MFnet)	eICU, mortality and Length of Stay
Ma et al. (2023) [3]	Patient similarity network, dynamic graph attention (PS - DGAT)	ICU mortality
Wang & Li (2025) [4]	Multimodal representation, medical ontology GCN, Transformer (TGAM)	MIMIC-III/IV, mortality among ICU sepsis patients
Che et al. (2018) [5]	GRU-D Recurrent modelling with missingness masks and time-decay information	Multivariate time-series prediction

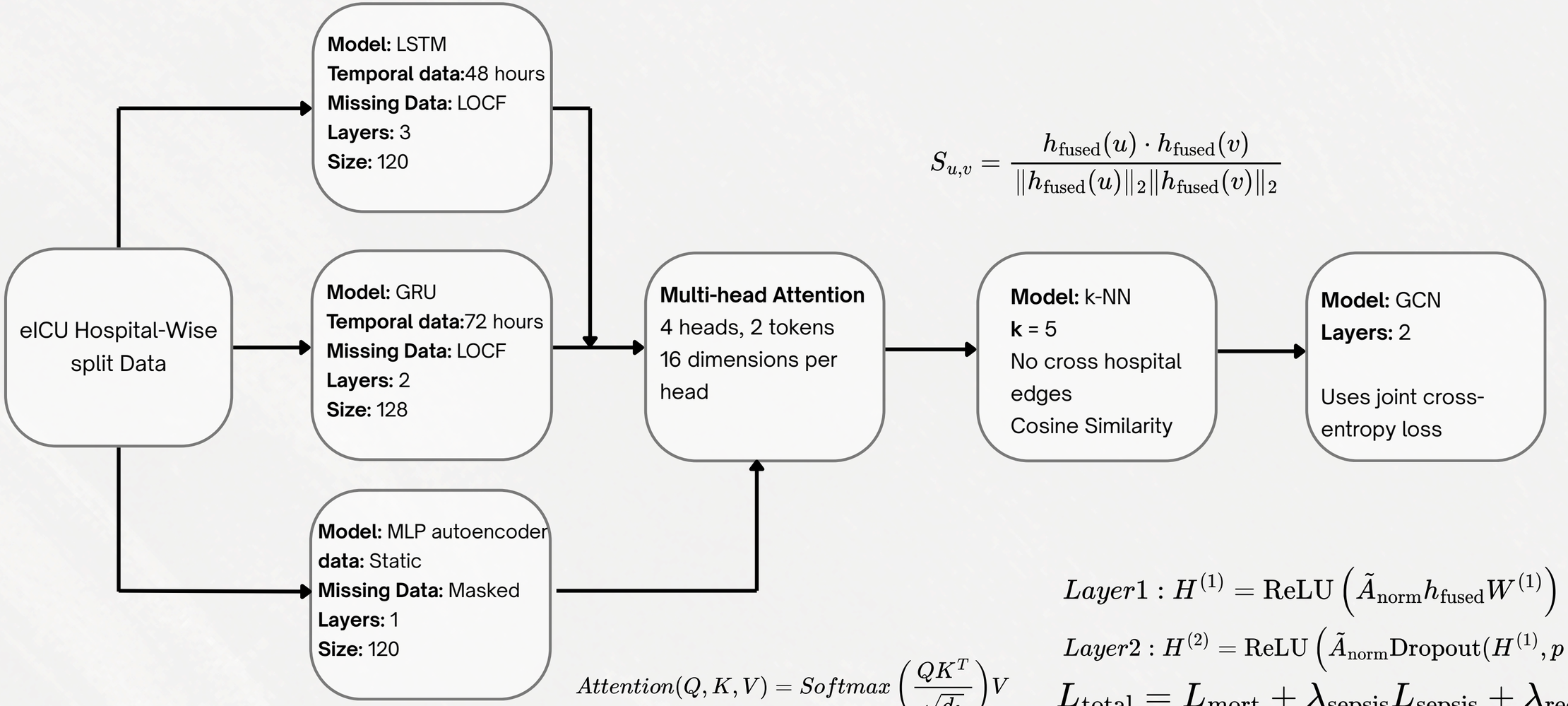
Problem Statement

Aim:
Predict Mortality and Sepsis on ICU patients based on data collected in the eICU-CRD

Doctors are overworked. This model aims to identify critical patients which can be cross checked by a professional to provide more care and early treatment before septic shocks.

Enforcing a strict hospital-level split alongside intra-hospital patient similarity graphs ensures the model's predictions reliably generalize to unseen healthcare institutions

Proposed Methodology



$$S_{u,v} = \frac{h_{\text{fused}}(u) \cdot h_{\text{fused}}(v)}{\|h_{\text{fused}}(u)\|_2 \|h_{\text{fused}}(v)\|_2}$$

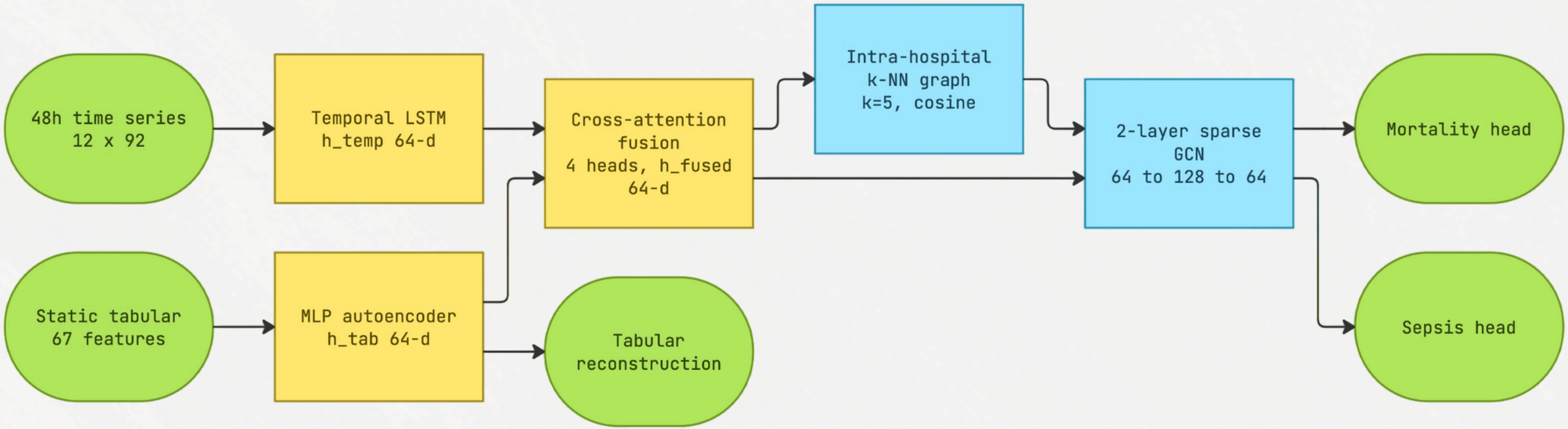
$$Attention(Q, K, V) = Softmax\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

$$Layer1 : H^{(1)} = \text{ReLU}\left(\tilde{A}_{\text{norm}} h_{\text{fused}} W^{(1)}\right)$$

$$Layer2 : H^{(2)} = \text{ReLU}\left(\tilde{A}_{\text{norm}} \text{Dropout}(H^{(1)}, p = 0.25) W^{(2)}\right)$$

$$L_{\text{total}} = L_{\text{mort}} + \lambda_{\text{sepsis}} L_{\text{sepsis}} + \lambda_{\text{recon}} L_{\text{recon}}$$

System Flowchart



Implementation

Subsystem / Stage	Module	Key Architectural Responsibilities
Stage 1: Temporal Encoder + Static Encoder	TemporalEncoder	92 input: features + masks; applies 128-d LSTM to output 64-d embedding 67 dimension input; MLP autoencoder encodes static features into a 64-d embeddings
Stage 2: Cross-Modal Fusion	AttentionFusion	Implements a 4-head self attention module with LayerNorm to yield a 64-d vector
Stage 3: Graph Convolution	GraphConvLayer & PatientGraphGNN	2-layer Kipf-Welling GCN over graphs; uses torch.sparse.mm for tensor multipliclations
Stage 4: Dual Outcome Heads	MortalityHead & SepsisHead	Parallel 2-layer MLP classification pathways (64 -> 32 -> 1) producing raw output logits
End-to-End Pipeline	MultimodalGNNPipeline	Master container reusing Stages 1–4 into a unified pipeline
Composite Objective	MultiTaskLoss	Calculates weighted BCE for mortality and sepsis combined with tabular MSE reconstruction loss

Results

Stage	Mortality AUROC	Mortality AUPRC	Sepsis AUROC	Sepsis AUPRC
Stage 1: Temporal Encoder	0.8474	0.4590	0.8668	0.6105
Stage 2: Tabular Encoder	0.8350	0.4410	0.7850	0.4120
Stage 3: Attention Fused	0.8793	0.5293	0.8850	0.5840
Stage 4: GCN	0.8706	0.4733	0.9131	0.6367

Metric	Mortality Prediction	Sepsis Prediction
AUPRC	0.4733	0.6367
Accuracy (%)	88.50	79.550
Recall (%)	64.79	88.34
Precision (%)	36.63	39.25
F1-Score	0.48	0.54

Conclusion and Future Work

The project investigated the use of Sequential and Graph based networks for predicting hospital mortality and Sepsis classification among ill patients using eICU–CRD. The GNN achieved a mortality AUROC of 0.8706 and 0.9131 for sepsis. These results demonstrate different performance across the two tasks with stronger discrimination for sepsis than for mortality.

This project can be extended in the future as follows:

- by adding more data modalities and a privacy preserving distributed learning. Federated learning framework would provide privacy for each hospital's data.
- FedAvg could be implemented initially, followed by personalized or adaptive aggregation.
- Explainability methods, including GNN Explainer and attention visualization
- Incorporate uncertainty estimation and model calibration.

References

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